

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CRYPTOGRAPHY AND NETWORK SECURITY
COURSE CODE: CSA51

COURSE OUTCOME COVERED

CO2: Analyze Public Key crypto system strategies using RSA and key Exchange for Encryption algorithm to strengthen information security. (BL4,BL5)

Scenario-Based : 20%

QUESTIONS:

Total Marks: 50

Annexure A

Code of Conduct:

I Sivakumar M(192521057) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Sivakumar M

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

CO2- 2- Scenario-based

1. Secure Data Transmission using Block Cipher

A company is transmitting repeated financial data blocks over a network. Initially, they used ECB mode and noticed patterns in ciphertext.

A Explain why this issue occurs.

Ans The company is using Electronic Code Book (ECB) mode to encrypt the financial data. In ECB mode, each plaintext block is encrypted independently using the same secret key. When two or more plaintext blocks contain identical data, they always produce the same ciphertext blocks. Since financial transactions often contain repeated values, repeated patterns appear in the ciphertext. These patterns can reveal information about the structure of the original data, making it easier for attackers to analyze the encrypted message. Although ECB encrypts the data, it does not hide patterns, so it is not suitable for transmitting sensitive financial information over a network.

B Suggest a more secure block cipher mode and justify your choice based on the scenario.

Ans A more secure block cipher mode for this scenario is Cipher Block Chaining (CBC) mode. In CBC mode, an Initialization Vector (IV) is used for the first block, and each plaintext block is combined with the previous ciphertext block before encryption. Because every ciphertext block depends on the previous one, identical plaintext blocks produce different ciphertext blocks. This removes visible patterns from the encrypted data and makes cryptanalysis much more difficult. As a result, CBC provides better confidentiality and stronger protection for repeated financial data. Therefore, CBC mode is a better choice than ECB for secure data transmission over a network.

2. Choosing an Encryption Algorithm for Banking System

A banking application needs high security and moderate performance for encrypting customer transactions. The available options are DES, 3-DES, and Blowfish.

A. Identify the most suitable algorithm for this scenario.

The most suitable encryption algorithm for the banking application is Blowfish. It provides strong security with a 128-bit key and offers faster encryption than DES and 3-DES. Since the banking system requires both high security and moderate performance, Blowfish is the best choice among the given algorithms.

B. Justify your choice based on security strength and efficiency.

Blowfish is preferred because it provides a good balance between security and performance. DES uses a **56-bit key**, which is considered weak and vulnerable to brute-force attacks. Although 3-DES improves security by using a **168-bit key**, it performs three rounds of DES encryption, making it much slower. Blowfish provides strong encryption with a **128-bit key** while requiring less encryption time than both DES and 3-DES. Its fast processing speed, low memory requirements, and strong resistance to attacks make it suitable for protecting customer transactions in a banking application. Therefore, Blowfish is the most appropriate algorithm for secure and efficient banking communication.

3. Key Exchange in an Unsecured Network

Two users want to establish a secure communication channel over a public network

using Diffie-Hellman key exchange. However, the network is vulnerable to attacks.

A. Explain how the key exchange works in this scenario.

Diffie-Hellman key exchange enables two users to establish a common secret key over a public network without transmitting the key directly. Both users first agree on a public prime number and a generator. Each user then selects a private key and computes a public key using the agreed values. They exchange these public keys over the network. Using their own private key and the received public key, both users independently calculate the same shared secret key. This shared secret key is then used to encrypt and decrypt the messages exchanged between them. Even though the public values are transmitted over the network, the private keys remain secret, making it difficult for an attacker to calculate the shared key.

B. Identify a possible security threat and suggest a solution to mitigate it.

A major security threat in Diffie-Hellman key exchange is the Man-in-the-Middle (MITM) attack. In this attack, an attacker intercepts the public keys exchanged between the two users and replaces them with their own public keys. As a result, each user unknowingly establishes a secret key with the attacker instead of with each other. The attacker can then decrypt, modify, and re-encrypt all messages without being detected. This threat can be prevented by using Digital Signatures, Digital Certificates, or a Public Key Infrastructure (PKI) to authenticate the exchanged public keys. Authentication ensures that the public keys belong to the intended users and prevents attackers from impersonating them.

4. Public Key Cryptography for Email Security

An organization plans to implement RSA for secure email communication.

A. Describe how RSA can be used for encryption and decryption in this scenario.

RSA is a public key cryptography algorithm that uses two different keys: a public key and a private key. For secure email communication, the sender encrypts the email using the receiver's public key. Since only the receiver possesses the corresponding private key, only the receiver can decrypt and read the email. This ensures the confidentiality of the email during transmission. RSA can also be used for digital signatures, where the sender signs the email using their private key, and the receiver verifies the signature using the sender's public key. This provides authentication, integrity, and non-repudiation.

B. Explain how key distribution and management play a role in maintaining security.

Key distribution and management are essential for maintaining the security of RSA. Public keys must be distributed securely so that users receive authentic keys belonging to the correct individuals. Digital certificates issued by a trusted Certificate Authority (CA) help verify the authenticity of public keys. Private keys must always remain secret and should be stored securely to prevent unauthorized access. Regular key updates, secure storage, and proper key management policies help protect the communication from attacks and ensure the continued security of the email system.

5. Cryptography for IoT Devices

A smart home system uses low-power IoT devices that need secure communication.

The designer must choose between RSA and Elliptic Curve Cryptography (ECC).

A. Analyze the constraints of IoT devices.

IoT devices usually have limited processing power, low memory capacity, restricted storage space, and limited battery life. They must perform encryption and decryption quickly while consuming very little power. Because of these resource limitations, cryptographic algorithms used in IoT devices should require less computation, less memory, and lower energy consumption without compromising security.

B. Recommend a suitable cryptographic approach and justify your answer.

Elliptic Curve Cryptography (ECC) is the most suitable cryptographic approach for IoT devices. ECC provides the same level of security as RSA while using much smaller key sizes. Smaller keys require less processing time, consume less memory, and reduce power consumption, making ECC ideal for low-power and resource-constrained devices. It also offers faster computations and efficient communication, which improves the overall performance of smart home systems. Therefore, ECC is the preferred choice for securing communication in IoT devices